

THE FUNDAMENTAL GROUP OF THE COMPLEMENT OF A PLANE CURVE WITH ONLY ORDINARY DOUBLE POINTS IS ABELIAN

PIERRE DELIGNE

INTRODUCTION

Theorem 1. Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a plane curve whose only singularities are ordinary double points with distinct tangents. Then the fundamental group of $\mathbb{P}^2(\mathbb{C}) \setminus C$ is abelian.

This theorem was stated by Zariski in 1929 [8]; see also [9, Chapter VIII]. His proof used Severi's theorem asserting that the space of irreducible plane curves of fixed degree with a prescribed number of double points is irreducible (*Vorlesungen über algebraische Geometrie*, 1921, Appendix F). It is still unknown whether Severi's assertion is true.

The group π_1 appearing in the theorem is the ordinary topological fundamental group. If one replaces $\mathbb{C}$ by an arbitrary algebraically closed field k of characteristic 0, the same statement holds for the algebraic fundamental group classifying finite étale coverings: by the Lefschetz principle one may reduce to $k = \mathbb{C}$, and then the algebraic fundamental group is the profinite completion of the ordinary one. In arbitrary characteristic one has the following variant.

Theorem 2 (Fulton [4]). Let k be an algebraically closed field, and let $C \subset \mathbb{P}_k^2$ be a plane curve whose only singularities are ordinary double points with distinct tangents. Then the tame fundamental group of $\mathbb{P}_k^2 \setminus C$ is abelian.

In other words, coverings in the sense of §0.2 of $\mathbb{P}_k^2$, ramified only along C and only tamely ramified there, are abelian. Recall that tameness can be tested after base change to the localizations of $\mathbb{P}_k^2$ at the generic points of the irreducible components of C ; these localizations are traits (that is, spectra of discrete valuation rings).

Assume now that k has characteristic $p > 0$. If C is as in Theorem 2, there always exists in the relative projective plane over $\text{Spec}(W(k))$ a relative normal-crossings divisor $\tilde{C}$ whose special fiber is C (see, for example, Mumford, appendix to Chapter VIII of [9]). This allows one to deduce Theorem 2 from Theorem 1 by specialization.

Once one knows that $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C)$ is abelian, homological arguments immediately yield its structure: if C is the union of irreducible curves of degrees $d_1, \dots, d_r$, then

$$\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C) \cong \mathbb{Z}^r / \mathbb{Z} \cdot (d_1, \dots, d_r).$$

In his proof of Theorem 2, Fulton uses on the one hand methods introduced by Abhyankar [1] to handle the special case where the irreducible components of C are smooth—methods that Serre explained very clearly in [7]—and on the other hand the following connectedness theorem. An analysis of Fulton's proof then allows the present writer to give a variant that yields Theorem 1.

Abhyankar and Fulton both exploit instances of the principle that, in algebraic geometry, anything sufficiently mobile is connected. Let $C \subset \mathbb{P}^2$ be as in Theorem 2, let D be an irreducible component of C , and let

$$f: X \rightarrow \mathbb{P}^2$$

be a covering ramified only along C and only tamely ramified there. Since $D \subset \mathbb{P}^2$ is very mobile, $f^{-1}(D)$ is connected. More precisely, D is ample in $\mathbb{P}^2$, hence $f^{-1}(D)$ is ample in X because f

Date: Séminaire Bourbaki, 32nd year, 1979/80, no. 543
English translation and typeset working edition.

is finite, and one applies Bertini. A local study of f shows moreover that if D is smooth then $(f^{-1}(D))_{\text{red}}$ is smooth, hence $f^{-1}(D)$ is irreducible.

In the general case let $g: \tilde{D} \rightarrow D$ be the normalization of D . The same local study shows that the reduced fiber product

$$(X \times_{\mathbb{P}^2} \tilde{D})_{\text{red}}$$

is smooth, hence is the normalization of $f^{-1}(D)$. If one can prove that it is connected, it follows that $f^{-1}(D)$ is irreducible, and one may then repeat the end of Abhyankar's argument as presented in [7].

Recall the definition of the fiber product: if

$$h: X \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2$$

is the product of $f: X \rightarrow \mathbb{P}^2$ and $g: \tilde{D} \rightarrow D \subset \mathbb{P}^2$, then $X \times_{\mathbb{P}^2} \tilde{D}$ is the inverse image $h^{-1}(\Delta)$ of the diagonal. Since $\mathbb{P}^2$ has many automorphisms, the diagonal $\Delta \subset \mathbb{P}^2 \times \mathbb{P}^2$ is very mobile, and therefore $h^{-1}(\Delta)$ is connected. This finishes the argument. More precisely, one applies to h the special case $r = 2$, $m = 2$ of the following theorem.

Theorem 3 (Fulton–Hansen [5]). Let

$$P = (\mathbb{P}^m)^r$$

be the product of r copies of projective m -space, let

$$\delta: \mathbb{P}^m \rightarrow P$$

be the diagonal map, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be a complete irreducible variety, and let

$$h: Z \rightarrow P.$$

Write $n = \dim h(Z)$ and $c = (r - 1)m$, the codimension of Δ in P . If $n - c \geq 1$, then $h^{-1}(\Delta)$ is connected.

This theorem has many other applications; for them I refer to [4] and to the paper of Gaffney and Lazarsfeld [5].

To end the introduction, let us explain on an example the method used to transform Fulton's proof of Theorem 2 into a proof of Theorem 1. The example is the case where C is smooth (hence irreducible).

Let U be a tubular neighborhood of C in $\mathbb{P}^2(\mathbb{C})$, put $U^* = U \setminus C$, and choose $u \in U^*$. For any covering

$$f: X \rightarrow \mathbb{P}^2(\mathbb{C})$$

ramified only along C , one has

$$(f^{-1}(C) \text{ connected}) \iff (f^{-1}(U) \text{ connected}) \iff (f^{-1}(U^*) \text{ connected}).$$

Thus the key result—that $f^{-1}(C)$ is connected for every such covering X —is equivalent to the following: for every finite quotient H of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$, the composite map

$$\pi_1(U^*, u) \xrightarrow{j_*} \pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u) \rightarrow H$$

is surjective. Morse theory proves more: the map

$$j_*: \pi_1(U^*, u) \rightarrow \pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$$

is itself surjective.

Once a Riemannian metric on $\mathbb{P}^2(\mathbb{C})$ has been chosen, one may, for sufficiently small $\varepsilon > 0$, identify U with the image under the exponential map of the radius- ε subbundle of the normal bundle of C in $\mathbb{P}^2(\mathbb{C})$. For sufficiently small ε , the exponential map identifies U^* with an oriented punctured-disc bundle over C , and hence there is an exact sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, u) \rightarrow \pi_1(C, \bar{u}) \rightarrow 1$$

(where $\bar{u}$ denotes the image of u in C). The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop γ going once positively around C . Since j_* is surjective, γ defines a central element of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$. The quotient of $\pi_1(\mathbb{P}^2(\mathbb{C}) \setminus C, u)$ by the normal subgroup generated by γ is

$$\pi_1(\mathbb{P}^2(\mathbb{C}), u) = \{e\},$$

so the theorem follows in this special case.

0. TERMINOLOGY

0.1. “Connected” means connected and nonempty. Likewise for “irreducible.”

0.2. A *covering* of a normal connected scheme S is a morphism $f: T \rightarrow S$ with T normal and connected, whose restriction to some dense open subset of T is étale.

0.3. Throughout the rest of the paper we work over $\mathbb{C}$. We simply write $\mathbb{P}^n$ for $\mathbb{P}^n(\mathbb{C})$. All schemes are assumed separated.

1. LEFSCHETZ– AND BERTINI-TYPE THEOREMS

The special case of Assertion 1.1 below in which $Z \subset \mathbb{P}^n$ is the complement of a hypersurface was proved by Hamm and Lê [6] by Morse theory. Our goal in this section is to deduce from it the variant, Theorem 1.6, of Theorem 3.

Assertion 1.1. Let $Z \subset \mathbb{P}^n$ be a nonempty Zariski-open subset. If H is a general hyperplane, the natural morphism

$$\pi_i(Z \cap H) \rightarrow \pi_i(Z)$$

is bijective for $i < n - 1$ and surjective for $i = n - 1$.

Assertion 1.1 is stated in [2], but Cheniot’s proof covers only the case where the complement of Z has codimension at least 2. Under that assumption, Z and $Z \cap H$ are simply connected, so it is harmless to work, as he does, with homology rather than homotopy groups. The case $i = 1$ may also be deduced from [6] or [2 bis].

Recall that the adjective “general,” when used of an element H of an irreducible algebraic variety S (here the dual projective space $\mathbb{P}^{n\vee}$), is a quantifier: it means that there exists a nonempty Zariski-open subset $S' \subset S$ such that the desired conclusion holds for every $H \in S'$.

1.2. Let

$$Y \subset \mathbb{P}^n \times \mathbb{P}^{n\vee}$$

be the set of pairs (z, H) such that $z \in Z$ and $z \in H$. Applying the isotopy theorem to the second projection

$$\text{pr}_2: Y \rightarrow \mathbb{P}^{n\vee}$$

gives a nonempty Zariski-open subset $U \subset \mathbb{P}^{n\vee}$ over which Y is a topologically locally trivial fibration. It is enough to take for U the set of hyperplanes transverse to the strata of an algebraic Whitney stratification of the complement of Z in $\mathbb{P}^n$.

Locally on U , after choosing a trivialization of this fibration, the inclusion of the fiber $Z \cap H = \text{pr}_2^{-1}(H)$ into Z appears as a map depending continuously on H from a fixed space into Z . Hence the conclusion of Assertion 1.1 either holds for every $H \in U$ or for none of them; in other words, Assertion 1.1 says precisely that it holds for all $H \in U$.

Suppose now that Z depends on parameters: let S be an irreducible scheme, let Z be an open subset of $S \times \mathbb{P}^n$, write

$$Z(s) = \text{pr}_2(\text{pr}_1^{-1}(\{s\})),$$

and assume each $Z(s)$ is nonempty. Let

$$Y \subset S \times \mathbb{P}^n \times \mathbb{P}^{n\vee}$$

be the set of triples (s, z, H) with $z \in Z(s)$ and $z \in H$. Applying the isotopy theorem to the projection of Y to $S \times \mathbb{P}^{n\vee}$ (respectively of Z to S) yields a nonempty Zariski-open subset $U \subset$

$S \times \mathbb{P}^{n^\vee}$ (respectively $V \subset S$) over which Y (respectively Z) is a topologically locally trivial fibration. As above, if $s \in V$ and $(s, H) \in U$, then the conclusion of Assertion 1.1 holds for $Z(s)$ and H . This parameterized variant will be used implicitly in the proof of Lemma 1.4.

I conjecture more generally the following.

Conjecture 1.3. Let Z be smooth and connected of dimension n , let

$$f: Z \rightarrow \mathbb{P}^N$$

be a morphism with finite fibers, let c be an integer, and let $L \subset \mathbb{P}^N$ be a linear subspace of codimension c . Set $n^* = n - c$. Choose a Riemannian metric on $\mathbb{P}^N$, and let

$$V(\varepsilon) = \{p \in \mathbb{P}^N \mid d(p, L) < \varepsilon\}.$$

Then, for sufficiently small ε , the natural morphism

$$\pi_i(f^{-1}(V(\varepsilon))) \rightarrow \pi_i(Z)$$

is bijective for $i < n^*$ and surjective for $i = n^*$.

For general L , the inclusion of $f^{-1}(L)$ in $f^{-1}(V(\varepsilon))$ is, for ε sufficiently small, a homotopy equivalence; thus Conjecture 1.3 is indeed a generalization of Assertion 1.1.

If one drops the assumption that f is quasi-finite and if the set of points of $\mathbb{P}^N$ over which the fiber of f has dimension k has dimension $\varphi(k)$, then one should redefine

$$n^* = 2 \dim Z - \sup(2k + \varphi(k) + \inf(\varphi(k), c - 1)) - 1.$$

The conjecture should moreover follow from a more precise statement to the effect that Z is obtained from $f^{-1}(V(\varepsilon))$ by attaching handles that kill spheres of dimension at least n^* .

We shall content ourselves with weaker results than Conjecture 1.3, deduced from the special case of Assertion 1.1 where Z is the complement of a hypersurface.

Lemma 1.4. Let Z be normal and connected, let

$$f: Z \rightarrow \mathbb{P}^N$$

be a morphism, let $n = \dim f(Z)$, and let c be an integer. Assume $n - c \geq 1$. If L is a general linear subspace of codimension c in $\mathbb{P}^N$, then $f^{-1}(L)$ is connected and, for every $z \in f^{-1}(L)$, the natural morphism

$$\pi_1(f^{-1}(L), z) \rightarrow \pi_1(Z, z)$$

is surjective.

One may replace Z by any nonempty Zariski-open subset Z' : for general L , the scheme $f^{-1}(L)$ is again normal, and Z' (respectively $Z' \cap f^{-1}(L)$) is obtained from Z (respectively $f^{-1}(L)$) by removing an algebraic subset with empty interior. In view of normality, this does not change π_0 and can only enlarge π_1 , as is seen from the commutative diagram

$$\begin{array}{ccc} \pi_1(Z' \cap f^{-1}(L), z) & \longrightarrow & \pi_1(f^{-1}(L), z) \\ \downarrow & & \downarrow \\ \pi_1(Z', z) & \longrightarrow & \pi_1(Z, z). \end{array}$$

In particular, we may and do assume that Z is smooth.

Shrinking Z , we may also assume that $f(Z)$ is locally closed and that Z is a topologically locally trivial fibration over $f(Z)$.

If $c = 1$ and f is dominant, then $f(Z)$ is a Zariski-open subset of $\mathbb{P}^N$ —after shrinking Z we may assume it is the complement of a hypersurface—and the lemma follows from Assertion 1.1 together with the long exact homotopy sequences for $Z \rightarrow f(Z)$ and $f^{-1}(L) \rightarrow L \cap f(Z)$.

Continue to assume $c = 1$. If f is not dominant, let A be a linear subspace of dimension $N - n - 1$, and let B be the projective space of linear subspaces of dimension $N - n$ containing A . Define

$$q: \mathbb{P}^N \setminus A \rightarrow B, \quad x \mapsto \text{the linear subspace containing } A \text{ and } x.$$

If A is disjoint from $f(Z)$, the restriction of q to $f(Z)$ is finite, hence dominant because $\dim f(Z) = \dim B$, and therefore $qf: Z \rightarrow B$ is dominant. Applying the already proved case of the lemma to qf and $c = 1$, we obtain the conclusion for a general hyperplane L among those containing A . Since A itself is only required to lie in a suitable dense open subset of a Grassmannian, the conclusion also holds for a general L in the usual sense (cf. 1.2).

The proof is completed by induction on c : a general linear subspace of codimension $c - 1$ inside a general hyperplane is a general linear subspace of codimension c (again cf. 1.2).

Lemma 1.5. Let Z , f , n , and c be as in Lemma 1.4. Every linear subspace L of codimension c in $\mathbb{P}^N$ admits a fundamental system of open neighborhoods V such that $f^{-1}(V)$ is connected and, for every $z \in f^{-1}(V)$, the natural morphism

$$\pi_1(f^{-1}(V), z) \rightarrow \pi_1(Z, z)$$

is surjective.

Let Gr be the Grassmannian of linear subspaces of codimension c in $\mathbb{P}^N$, let

$$Y \subset Z \times \text{Gr}$$

be the set of pairs (z, L) such that $f(z) \in L$, and let $G \subset \text{Gr}$ be a nonempty Zariski-open subset over which Y is a topologically locally trivial fibration. By 1.2, the conclusion of Lemma 1.4 holds for all $L \in G$.

For every open subset $U \subset \text{Gr}$, write U^* for the open subset of $\mathbb{P}^N$ that is the union of the $L \in U$. We claim that if U is connected, then Lemma 1.5 holds for $V = U^*$.

Indeed, if U is connected then $f^{-1}(U^*)$ is connected. The intersection $U' = U \cap G$ is connected, since it is obtained from U by deleting a triangulable subset of real codimension at least 2. By Lemma 1.4, the union $f^{-1}(U'^*)$ of the sets $f^{-1}(L)$ for $L \in U'$ is therefore connected. We now show that it is dense in $f^{-1}(U^*)$.

The image $f(Z)$ is not contained in the complement F of G^* , since $n - c \geq 0$ implies that the members of G meet $f(Z)$. If $z \in Z$, the image of any neighborhood of z cannot be contained in F , otherwise the image of all of Z would be, by analytic continuation. Thus if $z \in f^{-1}(U^*)$, there exist points z' arbitrarily close to z such that $f(z') \in G^*$. Among the $L \in \text{Gr}$ containing z' , the intersection with G is dense, because it is nonempty. If $z \in f^{-1}(L)$ with $L \in U$, this allows one to find z' as above, moreover lying in some $f^{-1}(L')$ with $L' \in G$ arbitrarily close to L . In particular, one may choose $L' \in U'$, so $z' \in f^{-1}(U'^*)$. This proves the density of $f^{-1}(U'^*)$ in $f^{-1}(U^*)$ and hence the connectedness of $f^{-1}(U^*)$.

Finally, if $L \in U'$, then $f^{-1}(L) \subset f^{-1}(U^*) \subset Z$. By Lemma 1.4, $\pi_1(f^{-1}(L))$ maps onto $\pi_1(Z)$, and therefore so does $\pi_1(f^{-1}(U^*))$.

We now derive a variant of Theorem 3.

Theorem 1.6. Let

$$P = (\mathbb{P}^m)^r$$

be the product of r copies of projective m -space, let

$$\delta: \mathbb{P}^m \rightarrow P$$

be the diagonal map, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be normal and connected, and let

$$h: Z \rightarrow P.$$

Write $n = \dim h(Z)$ and $c = (r - 1)m$, the codimension of Δ in P . If $n - c \geq 1$, then Δ admits a fundamental system of open neighborhoods V such that $h^{-1}(V)$ is connected. Whenever $h^{-1}(V)$

is connected and $z \in h^{-1}(V)$, the natural morphism

$$\pi_1(h^{-1}(V), z) \rightarrow \pi_1(V, z)$$

is surjective.

As in Lemma 1.4, we may shrink Z , and in particular we may assume it smooth.

Choose a family of hyperplanes $(H_\alpha)_{0 \leq \alpha \leq m}$ in $\mathbb{P}^m$ such that:

- (a) their intersection is empty, so in suitable projective coordinates they are given by $X_\alpha = 0$;
- (b) $h(Z)$ is contained in none of the inverse images $\text{pr}_i^{-1}(H_\alpha)$. After shrinking Z we may and do assume that $h(Z)$ is disjoint from every $\text{pr}_i^{-1}(H_\alpha)$;
- (c) $h(Z) \cap \Delta$ is not contained in the union of the $\delta(H_\alpha)$. Since Δ belongs to a connected family of cycles filling all of P , one has $h(Z) \cap \Delta \neq \emptyset$, and one can choose the H_α so that (c) holds.

Set

$$A_\alpha = \mathbb{P}^m \setminus H_\alpha, \quad B_\alpha = A_\alpha^r \subset P, \quad \Delta_\alpha = \delta(A_\alpha),$$

the diagonal in B_α . The spaces A_α and B_α are affine spaces and Δ_α is an affine subspace of B_α . The projective completion of A_α is $\mathbb{P}^m$; denote by $\overline{B}_\alpha$ the completion of B_α , and by $\overline{\Delta}_\alpha \subset \overline{B}_\alpha$ the closure of Δ_α .

Fix α , and for simplicity write $H, A, B, \Delta_A, \overline{B}$, and $\overline{\Delta}_A$ for $H_\alpha, A_\alpha, B_\alpha, \Delta_\alpha, \overline{B}_\alpha$, and $\overline{\Delta}_\alpha$.

Lemma 1.7. For every neighborhood U of Δ in P , there exists a neighborhood V of $\overline{\Delta}_A$ in $\overline{B}$ such that

$$V \cap B \subset U \cap B.$$

Choose on A a Hermitian vector-space structure compatible with its affine structure. A fundamental system of neighborhoods of Δ (respectively of $\overline{\Delta}_A$) has, upon intersection with B , the form

$$U \cap B = \left\{ (x_i) \left| \begin{array}{l} \text{if } |x_i| \leq R \text{ for some } i, \text{ then } |x_j - x_k| < \varepsilon \forall j, k, \\ \text{if all } |x_i| \geq R, \text{ then the angle between the lines } \mathbb{C}x_j, \mathbb{C}x_k \\ \text{is } < \varepsilon \forall j, k \end{array} \right. \right\},$$

$$V \cap B = \left\{ (x_i) \left| \begin{array}{l} \text{if } |x_i| \leq R \text{ for some } i, \text{ then } |x_j - x_k| < \varepsilon \forall j, k, \\ \text{if all } |x_i| \geq R, \text{ then } |x_j - x_k| \leq \varepsilon \inf_i |x_i| \forall j, k \end{array} \right. \right\}.$$

The lemma follows immediately.

In the case $r = m = 1$, the two compactifications $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ of $\mathbb{A}^2$ are dominated by a third one obtained by blowing up two boundary points; the original paper gives a schematic figure illustrating this and indicating the neighborhoods U and V .

We now return to the proof of Theorem 1.6. Apply Lemma 1.5 to

$$h: Z \rightarrow \overline{B}$$

and to the affine diagonal $\Delta_A \subset \overline{B}$. We find that $\overline{\Delta}_A$ has arbitrarily small open neighborhoods V_α such that

$$h^{-1}(V_\alpha) = h^{-1}(V_\alpha \cap B_\alpha)$$

is connected and has fundamental group mapping onto that of Z . By Lemma 1.7, given any neighborhood U of Δ in P , one can choose each V_α so that

$$V_\alpha \cap B_\alpha \subset U.$$

Then the union V of the opens $V_\alpha \cap B_\alpha$ is a neighborhood, arbitrarily small, of Δ in P . Hypothesis (b) on the hyperplanes H_α ensures that the intersection of the sets $V_\alpha \cap B_\alpha$ meets $h(Z)$. Hence

$$h^{-1}(V),$$

being a union of connected open sets $h^{-1}(V_\alpha)$ with nonempty mutual intersection, is connected. Since the π_1 of one of the $h^{-1}(V_\alpha)$ maps onto $\pi_1(Z)$, the same is true a fortiori for $\pi_1(h^{-1}(V))$, which completes the proof.

2. PROOF OF THEOREM 1

2.1. Let $C \subset \mathbb{P}^2$ be a reduced plane curve, let D be an irreducible component of C , let $C' = (C - D)$ be the union of the other components, let $\text{Sing}(C)$ be the set of singular points of C , and let

$$D^* = D - \text{Sing}(C).$$

Choose a base point $O \in \mathbb{P}^2 - C$, and let γ be a loop in $\mathbb{P}^2 - C$ of the following form: γ goes from O to a point p' near a point $p \in D^*$ by a path ch , winds once positively around D^* , and returns to O by ch^{-1} . Since D^* is connected, all loops of this type are mutually conjugate. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the normal subgroup they generate is

$$\pi_1(\mathbb{P}^2 - C').$$

Proposition 2.2. Assume that the singular points of C lying on D are ordinary double points with distinct tangents. Then γ is central in $\pi_1(\mathbb{P}^2 - C, O)$.

Let j be the inclusion of $\mathbb{P}^2 - C$ into $\mathbb{P}^2$, let $g: \tilde{D} \rightarrow D$ be the normalization of D , and apply Theorem 1.6 to

$$h = j \times g: (\mathbb{P}^2 - C) \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2.$$

To describe conveniently the inverse image of certain neighborhoods of the diagonal, we choose a Riemannian structure on $\mathbb{P}^2$ such that, in a neighborhood of every point $P \in D \cap \text{Sing}(C)$, the surface $\mathbb{P}^2$ is Euclidean and the two branches of C through P are orthogonal real 2-planes.

If $(u, v) \in (\mathbb{P}^2 - C) \times \tilde{D}$ lies in the inverse image of a sufficiently small neighborhood of the diagonal—that is, if u and v are sufficiently close—then u lies near D , and one can drop from u a short perpendicular to D . When u lies near $\text{Sing}(D)$ there are even two such perpendiculars, one for each branch. We choose the one whose foot $\bar{u}$ is close to v on $\tilde{D}$; then $\bar{u} \in D^*$.

Conversely, let $\varepsilon > 0$ be sufficiently small, let

$$q: U^* \rightarrow D^*$$

be the punctured-disc subbundle of the normal bundle to D^* in $\mathbb{P}^2$ consisting of nonzero normal vectors of length $< \varepsilon$, and let E be the set of pairs (x, v) with $x \in U^*$, $v \in \tilde{D}$, and

$$d(q(x), v) < \varepsilon$$

(the distance being computed on $\tilde{D}$). The projection

$$\text{pr}_1: E \rightarrow U^*$$

is a homotopy equivalence, with section given by the subspace of pairs $(x, q(x))$. The map

$$\text{exp}: (x, v) \mapsto (\text{exp}(x), v)$$

from E to $(\mathbb{P}^2 - C) \times \tilde{D}$ identifies E with the inverse image of a neighborhood of the diagonal, and as $\varepsilon \rightarrow 0$ one obtains the inverse images of a fundamental system of neighborhoods. Applying Theorem 1.6, we find that for any $O \in U^*$ the maps exp and q induce a surjection

$$(\text{exp}, q)_*: \pi_1(U^*, O) \rightarrow \pi_1(\mathbb{P}^2 - C, \text{exp}(O)) \times \pi_1(\tilde{D}, q(O)).$$

There is an exact homotopy sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, O) \rightarrow \pi_1(D^*, q(O)) \rightarrow 1.$$

The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop turning once positively around D^* . Since $(\text{exp}, q)_*$, and hence exp_* , is surjective, this loop determines a central element of $\pi_1(\mathbb{P}^2 - C, \text{exp}(O))$; this is precisely γ .

2.3. Proof of Theorem 1. For every irreducible component D of C , let $\gamma(D)$ be the corresponding central element of $\pi_1(\mathbb{P}^2 - C, O)$. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the central subgroup generated by the $\gamma(D)$ is

$$\pi_1(\mathbb{P}^2, O) = \{e\}.$$

Therefore $\pi_1(\mathbb{P}^2 - C, O)$ is abelian. □

REFERENCES

- [1] S. Abhyankar, *Tame coverings and fundamental groups of algebraic varieties*, Amer. J. Math. **81** (1959), 46–94.
- [2] D. Cheniot, *Un théorème du type de Lefschetz*, Ann. Inst. Fourier **25** (1975), no. 1, 195–213.
- [3] D. Cheniot, *Une démonstration du théorème de Zariski sur les sections hyperplanes d'une hypersurface projective et du théorème de Van Kampen sur le groupe fondamental du complémentaire d'une courbe plane*, Compositio Math. **27** (1973), no. 2, 141–158.
- [4] W. Fulton, *On the fundamental group of the complement of a node curve*, Ann. of Math. **111** (1980), no. 2, 407–409.
- [5] W. Fulton and J. Hansen, *A connectedness theorem for projective varieties, with applications to intersections and singularities of mappings*, Ann. of Math. **110** (1979), no. 1, 159–166.
- [6] T. Gaffney and R. Lazarsfeld, *On the ramification of branched coverings of $\mathbb{P}^n$* , Invent. Math. **59** (1980), no. 1, 53–58.
- [7] H. Hamm and Lê Dũng Tráng, *Un théorème de Zariski du type de Lefschetz*, Ann. Sci. É.N.S. **6** (1973), no. 3, 317–366.
- [8] J.-P. Serre, *Revêtements ramifiés du plan projectif (d'après S. Abhyankar)*, Séminaire Bourbaki, no. 204, May 1960.
- [9] O. Zariski, *On the problem of existence of algebraic functions of two variables possessing a given branch curve*, Amer. J. Math. **51** (1929), 305–328.
- [10] O. Zariski, *Algebraic Surfaces*, second supplemented edition, Ergebnisse der Mathematik und ihrer Grenzgebiete 61, Springer-Verlag.

Postscript (1 October 1980). I have just received a beautiful survey setting out the methods studied here, their variants, and their many applications:

W. Fulton and R. Lazarsfeld, *Connectivity and its applications in algebraic geometry*.

Pierre Deligne
 Institut des Hautes Études Scientifiques
 35 route de Chartres
 F-91440 Bures-sur-Yvette